

Paper – II: Dental Anatomy

Departmental objective

At the end of the course, the students should be able to

- Identify different teeth in different dentition
- Identify and describe different histological structures of Oro-dental Tissues
- Differentiate between normal and pathological state of oral structures.

List of competencies :

- Explain any tooth by its morphology
- Describe anatomy of any oral structures
- Mention the histological structures of tooth and associated oral structures
- Explain the clinical importance of teeth and oral tissues.

Distribution of teaching- learning hours

Lecture	Tutorial	Practical	Card exam	Total Teaching hours	Integrated teaching (Common)	Formative Exam		Summative Exam	
						Preparatory Leave	Exam Time	Preparatory Leave	Exam Time
99	146	27	18	290	10 hrs	20 days	42 days	30 days	30days

Teaching- learning methods, teaching aids and evaluation

Teaching Methods			Teaching Aids	In course evaluation
Large group	Small group	Self learning		
Lecture	Tutorial + Practical+ Demonstration	Assignments, Self-Study	Laptop, Computer, Multimedia projector, Slide Projector, OHP, White board, Study Models, Self-Jaw Model, Microscope, and Internet. 3D animation.	Continuous Evaluation (During delivering lecture) Assessment of learning Item exam Term 1 Term 2

Related Equipment's:

Multimedia projector, OHP, White board, Microscope, simulator, model, Internet

Professional Examination:

Marks Distribution of assessment of Professional Examination

Total Marks : 300

- Written : 100
- SOE : 100
- Practical : 100

Learning Objectives and Course Contents in Dental Anatomy

Oral Histology

Learning Objectives	Contents	Teaching Hours
Students should be able to – ● Define cell. ● Describe cellular organelles with functions. ● Explain the importance of intercellular junctions and basement membrane.	● Cell	L=2 hrs. T=2.5hrs
● Mention names of collagen producing cells and distributions of collagen in different tissues. ● Describe the process of collagen synthesis and degradation.	● Collagen	L=2 hr T= 1.5 hrs
● List the factors influencing hard tissue formation. ● Describe hard tissue genesis and degradation.	● Hard Tissue	L= 2 hrs T= 2 hrs
● Mention physical and chemical properties and types of dentine. ● Identify and describe dentinal tubules, incremental lines, predentine, peritubular dentine, intertubular dentine, interglobular dentine and granular layer of Tome's. ● Describe age changes and sensitivity theories of dentine. ● Mention the clinical importance of dentine.	● Dentine	L= 3 hrs T= 4.5 hrs P= 2 hrs
● State composition and function of pulp. ● Describe different histological zones of pulp. ● Outline the distribution of nerves and vessels in pulp. ● Describe pulp stones, age changes of pulp and its clinical importance.	● Pulp	L= 2 hrs T= 2 hrs P= 2 hrs
● Mention physical and chemical properties of enamel. ● State the orientation of enamel rods and its clinical importance. ● Identify and describe the incremental lines, enamel tufts, enamel lamellae, enamel spindles, Hunter-schreger bands, dentino-enamel junction, cement-enamel junction. ● Describe surface features of enamel, age changes and clinical significance.	● Enamel	L=3 hrs T= 4.5 hrs P= 3 hrs
● Describe different types of cementum,, their distribution and function. ● State briefly afibrillarcementum, intermediate cementum . ● Mention the process of repair of root surface.	● Cementum	L= 2 hrs T= 2hrs P= 2 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – ● State different cells and fibers of periodontal ligament and their function. ● Describe different sources of nutrition of periodontal ligament. ● Differentiate between diseased and healthy periodontal tissue. ● Mention the clinical importance of periodontal ligament.	● The periodontal Ligament	L= 2 hrs T= 2.5 hrs P= 2 hrs
● Mention types and function of oral mucosa. ● Describe structural variations of oral mucosa and gingiva and their functional importance. ● Mention normal histological features of oral mucosa. ● State dento-gingival junction, muco-cunateous junction and alveolo-gingival junction. ● Describe different papillae of tongue, keratinization, non-keratinocytes of oral epithelium.	● Oral mucosa & Gingiva	L= 3 hrs T= 3 hrs P= 5 hrs
● Describe alveolar bone, ● Explain clinical importance of alveolar bone.	● Alveolar bone	L= 2 hrs T= 2.5 hrs P= 1.5 hrs
● Define joints and synovial joints. ● Describe synovial fluid and synovial membranes. ● Mention the characteristics of synovial joint and unusual features of TMJ (temporo-mandibular) as a synovial joints. ● Describe different structures related to TMJ. ● Describe the histological aspect of TMJ. ● Mention clinical symptoms related to TMJ disorder.	● Temporo-mandibular joint	L= 2 hrs T= 4 hrs
● Define saliva and mention its composition and function and relation to dental carries. ● Classify salivary glands. ● Describe the formation and secretion of saliva. ● Mention the distribution of salivary glands. ● Differentiate histologically serous and mucous salivary glands. ● Describe the ductal systems of salivary glands. ● Describe myoepithelial cell. ● Explain the clinical importance of salivary gland and some pathological condition related to salivary gland.	● Salivary Glands	L= 3 hrs T= 5.5 hrs P= 3.5 hrs

Dental Morphology

Learning Objectives	Contents	Teaching Hours
Students should be able to- - Describe types of dentition, dental formula and notation. - Shows various anatomical landmarks of teeth. - Mention the chronology of deciduous and permanent teeth.	Introduction	L= 2 hrs T= 4 hrs
- Identify central incisors and their normal landmarks. - Mention their normal relationship in dental arch and clinical importance.	Central Incisors	L= 2 hrs T= 3 hrs
- Identify lateral incisors and their normal landmarks. - Show their normal relationship of lateral incisors in dental arch and mention its clinical importance.	Lateral Incisors	L= 2 hrs T= 3 hrs
- Identify canines and their normal landmarks. - Mention their normal relationship and clinical importance.	Canines	L= 2 hrs T= 3 hrs
- Identify premolars and their normal landmarks. - Mention their normal relationship and clinical importance.	Premolars	L= 2 hrs T=4 hrs
- Identify maxillary molars and their normal landmarks. - Mention their normal relationship and clinical importance.	Maxillary Molars	L= 3 hrs T= 5.5 hrs
- Identify mandibular molars and their normal landmarks. - Mention their normal relationship and clinical importance.	Mandibular Molars	L= 4 hrs T= 6.5 hrs
- Identify deciduous teeth. - Differentiate deciduous teeth from permanent teeth.	Deciduous Teeth	L= 3 hrs T= 5 hrs

Applied Anatomy

Learning Objectives	Contents	Teaching Hours
Students should be able to- ● Mention the boundary and relation of maxillary sinus. ● State the lining, nerves and vascular supply of maxillary sinus. ● Describe clinical importance of maxillary sinus.	● Maxillary Sinus	● L= 2 hrs ● T= 1.5 hrs
● Mention vascular supply, lymphatic drainage and innervations of oro-dental tissues. ● Describe the route of transmission of pathological lesion from oral cavity to other locations.	● Vascular Supply ● Lymphatic Drainage ● Innervations of Oro-Dental tissues.	● L= 4 hrs ● T= 5 hrs

Oral Physiology

Learning Objectives	Contents	Teaching Hours
● Students will be able- ● Explain normal occlusion. ● State the ideal criteria and factors that influence normal occlusion. ● Define malocclusion. ● Mention types of malocclusion.	● Occlusion	● L= 3 hrs ● T= 3.5 hrs
● State different masticatory movements. ● Describe the mechanism of mastication. ● Describe masticatory muscles, their attachments and function in mastication. ● Describe neural control of mastication.	● Mastication	● L= 2 hrs ● T= 2 hrs
● Describe different types of physiological tooth movement. ● Describe the mechanisms and pattern of shedding. ● State mechanisms and theories of tooth eruption. ● Mention the molecular determinants of tooth eruption. ● chronology and pattern of tooth eruption and its clinical significance.	● Eruption	● L= 2 hrs ● T= 3 hrs
● Define shedding. ● Describe the mechanism and pattern of shedding. ● Describe the histology of odontoclast cell. ● Describe the gubernacular cord and canal. ● Mention the clinical importance of shedding.	● Shedding	● L= 2 hrs ● T= 3 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – ● Mention the phases of swallowing. ● State the difference between mature and immature swallowing. ● Describe the mechanisms of control of swallowing., dysphasia and clinical consideration.	● Swallowing	● L= 3 hrs ● T= 4.5 hrs
● Describe phonation and neural basis of language. ● Classify sounds. ● State the clinical significance related to speech.	● Speech	● L= 3 hrs ● T= 4.5 hrs
● Define pain. ● Describe the physiology of pain in oral region and its pathway to CNS.	● Pain & pain Pathways	● L= 2 hrs ● T=1.5 hrs
● Describe taste buds and taste receptors. ● State the role of saliva in taste perception. ● Describe the pathways of taste sensation.	● Taste & Taste Pathway	● L= 1 hr ● T= 1.5 hrs
● Describe the mechanisms of sucking and its clinical consideration.	● Sucking	● L= 1 hrs ● T= 4.5 hrs

Embryology

Learning Objectives	Contents	Teaching Hours
Students should be able to- ● Mention germ cell formation and fertilization. ● State the formation of three-layered embryo, formation of neural tube and neural crest. ● Mention the derivatives of different germ layers and neural crest.	● Basic Embryology	L= 2 hrs T= 6 hrs
● State briefly the formation of pharyngeal arches and pouches. ● Describe the derivatives of pharyngeal arches and pouches.	● Pharyngeal Arch, clefts & pouches	L= 1 hr T= 1.5 hrs
● Mention the different prominences involving face development. ● State the congenital defects during face development.	● Development of face & ● Congenital defects	L= 2 hrs T= 3 hrs
● Outline the development of maxillae, mandible, palate, tongue and lips. ● State the congenital defects during their formation.	● The Development of Jaws, ● palate, tongue & Lips	L= 5 hrs T= 6 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to- - Define primary epithelial band, dental lamina and successional lamina. - Describe the bud, cap and bell stages of tooth development. - Describe the physiological process during tooth development.	Early Development of teeth	L= 4 hrs T= 6 hrs P= 6 hrs
- Describe the formation of enamel, dentine, cementum, pulp, periodontal ligament, salivary glands and Temporo-mandibular joint. - State the defects that may occur during their development and their clinical significance.	Amelogenesis, Dentinogenesis, Cementogenesis, Development of pulp and periodontal ligament, Salivary Gland, TMJ	L= 8 hrs T= 12 hrs

Comparative Dental Anatomy

Learning Objectives	Contents	Teaching Hours
Students should be able to- - Outline the characteristic of human dentition,. - Describe the different mode of attachment of teeth and evolution of tooth in vertebrates.	Characteristics of human dentition, Types of Tooth attachment.	L= 2 hrs T= 3 hrs
- Describe the variations of the dental tissues into different species. - State the characteristic and histology of rodent incisors.	Comparative anatomy of dental tissues and Rodent incisor.	L= 2 hrs T= 3 hrs

NB:

- Details of salivary glands , development of Face, tongue, palate, tooth, oral cavity, salivary glands, thyroid gland, parathyroid gland, pharyngeal arches, pouches and cleft are included both in anatomy and dental anatomy. So it is suggested to teach and assess these topics in Dental Anatomy
- Detail of cell biology and general embryology are included both in anatomy and dental anatomy.